

The $\chi(P)$ carrier-richness link is bypassed: the WEIGHTED Lemma A (R-027, T7)

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The operator verdict named two residuals; this note addresses Residual 1, the carrier-richness link $\chi(P) \lesssim T'$ from the additive energy to the physical STEP-5B floor. The conclusion is that it is BYPASSED – a proved (T7) weighted bridge gives the floor bound directly in terms of T' . Scope: B1's lattice T6 scope; any amplitudes; the $I = 2 \times 10^{-3}$ endpoint. No tier action.

2. The weighted Lemma A is the $\chi(P)$ -free link (with proof sketch)

The STEP-5B floor requires $\sum_t w_t^2 \leq K(\lambda'I)^2$ with $w_t = \sum_{u+v=t} c_u c_v$. Write $r(t) = \#\{(u, v) \in M \times M : u + v = t\}$ for the pair multiplicity of the transfer t . By Cauchy–Schwarz on the $r(t)$ admissible pairs,

$$w_t^2 = \left(\sum_{u+v=t} c_u c_v \right)^2 \leq r(t) \sum_{u+v=t} |c_u|^2 |c_v|^2.$$

For $t \neq 0$ the pairs (u, v) with $u + v = t$ have both summands on the sum-level circle $C_t = \{x : x \cdot t = |t|^2/2\}$, so $r(t) \leq \#(M \cap C_t) \leq T'(Q) = \max_{t \neq 0} \#(M \cap C_t)$. Summing and using $\sum_t \sum_{u+v=t} |c_u|^2 |c_v|^2 = (\sum_u |c_u|^2)^2 = \|c\|_2^4$,

$$\sum_{t \neq 0} w_t^2 \leq T'(Q) \sum_{t \neq 0} \sum_{u+v=t} |c_u|^2 |c_v|^2 \leq T'(Q) \|c\|_2^4,$$

which is the $t \neq 0$ part of the WEIGHTED Lemma A $\sum_t w_t^2 \leq (1 + T'(Q)) \|c\|_2^4$ (R-027, dr2-step5b-integration v1.1, T7).¹ Since $w_0 = I$,

$$K_{\text{floor}}(c) = \frac{\sum_{t \neq 0} w_t^2}{I^2} \leq T'(Q) \quad \text{for ANY amplitudes.}$$

The additive-energy $\rightarrow$ STEP-5B-floor connection is THIS weighted bridge. The carrier-richness $\chi(P)$ is the DUAL extraction-route incidence quantity; the integration route uses T' directly and does NOT invoke $\chi(P)$. (Verified: over a random-amplitude subset of 115 configs $\times$ 3 random amplitude profiles, $K_{\text{floor}}(c) - T' \leq 0$; the full admissible T' -pin set is 130 configs, T-014/T-015.)

¹The $t = 0$ channel is split off separately: for a real field with the antipodal pairing $c_{-u} = \overline{c_u}$ and the normalisation $\|c\|_2^2 = I$, $w_0 = \sum_u c_u c_{-u} = \sum_u |c_u|^2 = \|c\|_2^2 = I$ exactly, contributing the “1” in $(1 + T')$; the floor uses only the $t \neq 0$ part.

3. The pin: physical floor ≤ 13 for any amplitudes

The admissible pin (res-endpoint-admissible-pin) gives $\max_Q T' = 13$ over the admissible crystallographic-shell class (130 configs). Hence, by the weighted Lemma A,

$$K_{\text{floor}}(c) \leq T'(Q) \leq 13 < 20.5 \text{ (RES-1)} < 26.2 \text{ (RES-5)} \quad \text{for ANY amplitudes,}$$

i.e. the PHYSICAL STEP-5B floor (not merely the uniform-amplitude additive energy) is pinned below both thresholds, $\chi(P)$ -free.

4. Verdict: Residual 1 closed within scope

Residual 1 ($\chi(P)$ link) is BYPASSED. The physical STEP-5B floor is bounded by T' directly via the weighted Lemma A (R-027, T7), and pinned ≤ 13 by the admissible scan – for any amplitudes. So the “EXACT additive-energy pin” upgrades to a PHYSICAL STEP-5B floor pin within B1’s lattice scope, at the grade

$$\{\text{weighted Lemma A T7}\} + \{\text{exact } T' \text{ pin} \leq 13\}.$$

Remaining (honest): (a) the competitor-class-definition – that competitors are exactly crystallographic-shell subsets up to n_{pack} (an operator/modeling confirmation, the R-027 residual (ii)); (b) Residual 2 – the off-diagonal worst-direction operator norm for H-diag (T-018). The class-wide $T' \leq C_\varepsilon R^\varepsilon$ is R-026 (T7 modulo textbook NT), but T' is PINNED EXACTLY ≤ 13 over the admissible range, so that caveat is moot within scope. No tier flip: B1/B2 stay T6 CONDITIONAL on {H-LAYER}.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*Is the weighted Lemma A really T7 and really $\chi(P)$ -free?*” The proof sketch (§2) is an elementary Cauchy–Schwarz / pair-multiplicity bound; its statement is in T' , not $\chi(P)$; the $\chi(P)$ extraction route is a SEPARATE (dual) approach. The numerical check confirms $K_{\text{floor}}(c) \leq T'$ for random amplitudes. So Residual 1 is genuinely bypassed, not relabelled.
- (β) “*So is the floor a theorem now?*” Within B1’s lattice scope, the floor pin rests on weighted Lemma A T7 + exact $T' \leq 13$ pin – high grade, but the competitor-class-definition is an operator/modeling item and the threshold (20.5/26.2) is estimate-grade. So STRONG EVIDENCE for the physical-floor closure, not an unconditional theorem.
- (γ) “*Does this discharge H-LAYER?*” No – it closes Residual 1 (the $\chi(P)$ link). Residual 2 (operator norm, T-018) and the Prop-A isotropy core (T-016) remain. B1/B2 T6.

Quantitative sanity check (mandatory): *weighted Lemma A* $\max(K_{\text{floor}}(c) - T') = 0$ over 115 configs $\times 3$ random profiles (full pin set 130 configs); *pin* $\max T' = 13 < 20.5 < 26.2$; $t=0$ $w_0 = \|c\|_2^2 = I$ exactly (antipodal real field); *grade* T7 weighted bridge + exact T' pin.

6. Result footer

Result ID: B1-RH-ENUM / $\chi(P)$ link bypass (Residual 1; v1.1 proof-sketch re-issue)

Precise statement: The weighted Lemma A (R-027, T7) gives $\sum_t w_t^2 \leq (1+T')||c||_2^4$ for any amplitudes [proof: Cauchy-Schwarz $w_t^2 \leq r(t) \sum |c_u|^2 |c_v|^2$, $r(t) \leq T'$ for $t \neq 0$; $w_0=I$], i.e. $K_{\text{floor}}(c) \leq T'$ -- the additive-energy $\rightarrow$ STEP-5B-floor link, $\chi(P)$ -free ($\chi(P)$ is the dual extraction-route quantity). With the admissible pin max $T'=13$, the physical floor is $K_{\text{floor}} \leq T' \leq 13 < 20.5 < 26.2$ for ANY amplitudes. Residual 1 ($\chi(P)$ link) is BYPASSED.

Scope: B1 lattice T6 scope; any amplitudes; $I=2e-3$. STRONG EVIDENCE.

Dependencies: dr2-step5b-integration v1.1 (R-027 weighted Lemma A, T7); res-endpoint-admissible-pin (max $T'=13$, 130 configs); res5-dr2-kappa-bound v1.2; res5-arc-consolidation.

Evidence grade: ANALYTIC (weighted Lemma A proof sketch, R-027 T7) + EXECUTED (res5_017_chi_bypass.py 3/3, 115-config random-amplitude subset).

Reproduction command: python codes/vacuum/res5_017_chi_bypass.py

Expected output: $K_{\text{floor}}(c)-T' \leq 0$ (any amplitudes); max $T'=13 < 20.5 < 26.2$; $\chi(P)$ bypassed; 3/3 PASS.

Falsification gate: An amplitude profile with $K_{\text{floor}}(c) > T'$ (would break the weighted Lemma A); or an admissible config with $T' > 13$.

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}. Residual 1 ($\chi(P)$ link) CLOSED/bypassed within scope; physical-floor pin via R-027 T7 + T' pin. Remaining: class-definition + Residual 2 (T-018).

No-overclaim statement: STRONG EVIDENCE, not an unconditional theorem: {weighted Lemma A T7}+{exact T' pin} close the physical floor within scope, but the competitor-class-definition is an operator item and the threshold is estimate-grade. $\chi(P)$ is genuinely bypassed (the weighted bridge is in T' , not $\chi(P)$). B1/B2 T6 unchanged.

Next required action: T-018 (off-diagonal operator norm) and the Prop-A isotropy core (T-016) are the remaining H-LAYER residuals; the $\chi(P)$ link is removed.